

# **Abstract**

## **FoodLens**

**AI-Powered Ingredient Scanner and Health Risk Scoring System**

**Group -11**

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# **FoodLens : AI-Powered Ingredient Scanner and Health Risk Scoring System**

*for Packaged Food Products — MCA Major Project Synopsis*

## **Abstract**

Packaged and processed food consumption has risen sharply across urban and semi-urban India, alongside a parallel rise in diet-related health concerns such as obesity, diabetes, and hypertension, many linked to hidden sugars, unhealthy fats, sodium, and synthetic additives. Consumers have limited practical means of interpreting dense, technical ingredient labels at the point of purchase. Existing global platforms such as Open Food Facts and apps like Yuka offer ingredient scoring, but suffer from poor coverage of regional Indian products and generic, one-size-fits-all scores that ignore individual health conditions.

This project proposes an AI-assisted mobile/web-based Ingredient Scanner and Health Risk Scoring System. Users scan a barcode or, where no match exists, photograph the ingredient label directly. The system extracts, cleans, and interprets the ingredient list, computes a transparent, explainable health risk score, and personalizes it to the user's health conditions and allergies. It further extends into a fuller dietary-assistance platform through comparison tools, consumption tracking, community-sourced regional product data, and accessibility features.

The system follows a hybrid architecture combining deterministic, rule-based scoring, for accuracy and defensibility of health claims traceable to sources such as FSSAI and WHO/ICMR, with AI/NLP used specifically for OCR cleanup, ingredient matching, and natural-language explanation generation. This separation of what is claimed (rule-based) from how it is communicated (AI-phrased) keeps health assessments consistent and scientifically grounded while benefiting from LLM fluency in user communication.

The system incorporates a set of user-facing and community-oriented features designed to increase real-world utility and encourage long-term engagement rather than one-off use, including side-by-side product comparison, personal consumption tracking with exportable and scheduled reports, community-sourced regional product data verification, simple sharing, an administrative and analytics layer, and accessibility features such as multi-language support and voice playback. Together these are intended to transform the application from a single-purpose scanning utility into a continuously improving, personalized dietary assistance platform.

## **Objective**

To design a system that: retrieves ingredient data via barcode lookup; extracts ingredients from labels via OCR for regional products missing from global databases; scores health impact using a transparent, source-cited, rule-based methodology; personalizes scores to each user's health profile and allergies; uses LLMs only for OCR cleanup and plain-language explanation of pre-verified results; and extends into comparison, tracking, community contribution, and accessibility features — built as a modular system dividable across a four-member team.

## Module and Team Division (4 Members) with Technologies Used

The nine underlying system modules — input acquisition, barcode lookup, OCR and text extraction, NLP ingredient parsing, the ingredient risk database, the scoring engine, personalization, explanation generation, and the user interface/dashboard — are grouped into four coherent vertical slices below, so that each team member owns an end-to-end capability rather than an isolated fragment, with clearly defined integration points between members.

### Member 1 (Amal Reghunath) — Input Acquisition & Barcode Lookup (Frontend-Data Entry Lead)

- **Modules:** Builds the camera/scan interface for barcode, QR, and label-image capture; integrates the Open Food Facts API for barcode-matched products; routes unmatched scans to the OCR pipeline; handles API/error parsing.
- **Technologies:** React Native / React + ZXing-js, Open Food Facts REST API, Axios/Fetch, Redux/Context API.

### Member 2 (Akhil Alex) — OCR & NLP Ingredient Parsing (AI/Text Processing Lead)

- **Modules:** Runs OCR on label images with preprocessing (cropping, contrast, deskew) for regional packaging; displays a confidence score; cleans and structures text via NLP and alias resolution (e.g., E211 → Sodium Benzoate); uses an LLM narrowly for OCR-typo correction.
- **Technologies:** Tesseract OCR / Google Cloud Vision API, OpenCV, spaCy/regex, Gemini/GPT API (text cleanup only), Python Flask microservice.

### Member 3 (Lekshmi A Nair) — Ingredient Database & Scoring Engine (Backend/Data Lead)

- **Modules:** Builds and populates the ingredient risk database (AI-assisted draft, manually verified against FSSAI/WHO); designs the schema; implements position-weighted scoring and normalization; implements personalization multipliers and allergen overrides; manages user health profiles.
- **Technologies:** MySQL, Django REST Framework / Flask, Python (scoring logic), Pandas.

### Member 4 (Mithul Jacob Manoj) — Explanation Generation & UI/Dashboard (Frontend-Output & Integration Lead)

- **Modules:** Builds the results screen (traffic-light score, breakdown, warnings); integrates constrained LLM explanation generation; builds scan history dashboard; owns overall integration of all four members' modules into one pipeline.
- **Technologies:** React / React Native, Chart.js/Recharts, Gemini/GPT API, Tailwind CSS, Firebase/MySQL.

Integration Flow: Member 1 (Scan) → Member 2 (OCR/NLP, if no barcode match) → Member 3 (Scoring & Personalization) → Member 4 (Explanation & Display).

## Extended Features

Beyond the core scan-to-score pipeline, the following features broaden the system's practical value and demonstrate additional depth and breadth of functionality:

- **Product Comparison and Recommendations** — allows users to compare two scanned products side by side, receive a clear "healthier choice" recommendation, view alternative product suggestions with higher health scores from the existing database, and get recipe/meal suggestions built around healthier ingredients identified during scanning.
- **Consumption Tracking and Reporting** — maintains a daily/weekly consumption tracker summarizing trends such as total sugar or sodium intake from scan history, generates exportable PDF scan reports suitable for sharing with a doctor or dietitian, sends scheduled personalized health reports via email, and issues notifications based on consumption patterns.
- **Community-Sourced Product Database** — lets users submit ingredient data for unrecognized regional or local products that global databases typically miss, with OCR pre-filling a draft ingredient list for the user to review; submissions are routed through admin review or automatic multi-user consensus approval before being added to the verified database.
- **Sharing** — generates a shareable QR code or link for a scanned product's health score, allowing results to be easily shared with family, friends, or healthcare providers.
- **Administration and Analytics** — provides an admin dashboard for reviewing and approving community submissions, and an aggregate analytics view highlighting the most commonly flagged ingredients across all users over a given period.
- **Accessibility** — includes a multi-language interface toggle for regional language support, text-to-speech playback of score explanations for visually impaired or elderly users, and dark mode/general accessibility settings.

## Expected Outcome

A working prototype capable of scanning packaged food products, with particular strength on Indian regional items underserved by existing global databases, generating an accurate, personalized health score with a clear explanation, and supporting continuous improvement through community contribution — helping users, especially those managing chronic conditions or allergies, make more informed dietary decisions.

## Keywords

OCR, NLP, Health Informatics, Ingredient Risk Scoring, Personalized Nutrition, Food Safety, Rule-Based Systems, LLM-Assisted Explanation Generation, Crowdsourced Data Verification.